Supplementary Information:
The Lonely Runner Problem on Pair-Sum Lattices

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Contents

1 The verification record of the scaling-closure derivation

Every number above is either an exhaustive census with two independent decision procedures, or a sample with its size and interval, or a proved statement with its machine check; the records, in the order of appearance: the pool censuses (the pruned-cascade and the flattened-popcount procedures agree exactly at $p = 17/19/23$; an independent pure-enumeration brute force agrees on 21/21 sampled tuples across $p = 17/19/23/31$, covering both admissible and inadmissible branches; the $T = 9$ stride samples carry full decisions, and the $m = 6$ code path shares the cascade code brute-force-verified at four $m = 5$ primes, with the external consistency check the agreement of the projected pools with the independently measured pool scale); the zone-resolved γ table (764 pairs, five committed cells, the identical seed reproducing the committed family selections); the chain-class machine checks (123,240 solutions, 0 violations of the sum identity or the reach bound; the count reconciliations 131,520 against 123,240 and 10,080 against 8,880); the reflection fiber (survival exactly 1 at every $\bar{\lambda} = 1$ pair, every family, every cell); the renormalization-orbit law (machine-checked set-by-set at $p = 17$ and $p = 19$); the volume census and its ground truth (9/9 brute-force keys; the admissibility reproductions 840/840 and 960/1,680; the order-independence of volume across all 24 orderings); the essential-arc, budget-identity, and cap checks on the probed families; and the cascade depths at the committed cells (medians 2, maximum 4); and the character-system record of §**??** (the rigidity census over 1,678 admissible keys with every key inside the committed census scope matching it exactly, 1,654/1,654, 0 mismatches; the per-assignment renormalization bijection, 801/801 images injective with per-key counts matching under the explicit σ; the falsifications of V' and V''; the character-system aggregate invariance on both full top orbits; the chain controls 8,880/4,680/29,760 reproduced by the same code path). The transfer-matrix route's record: the generic dichotomy and trace fingerprints (1,500 systems, 12,000 comparisons, zero mismatches); the 66 homogenized pair systems on the six committed families with the closed-form orbit inventories verified by full traversal and the finite-horizon survival counts verified by direct simulation (zero mismatches); the loop-composition identity (algebraic and numeric, 500 points per family); the per-step equality of the automaton's survivor counts with the committed μ-space cascade convention (zero mismatches, all steps, all families; the committed chain-family depth reproduced); the end-to-end fiber-covering ground truth (2,800/2,800); the reflection pairs (survival fraction exactly 1, all six families, all three primes); and the reproduction of the committed volume margins (876/108/186 against $95\times$ /1,207 $\times$ /3,803$\times$). The three-rung depth record of Table **??** is calibration from the earlier cascade runs and is labeled as such wherever it is used.